Little Explorers · Class 7

Plant Water Alarm

A gadget that tells us when our plant is thirsty!

Soil sensor | Wet or dry? | Beep alarm



Did you know?
Some farms use soil sensors so they only water plants when they really need it. That saves lots of water!

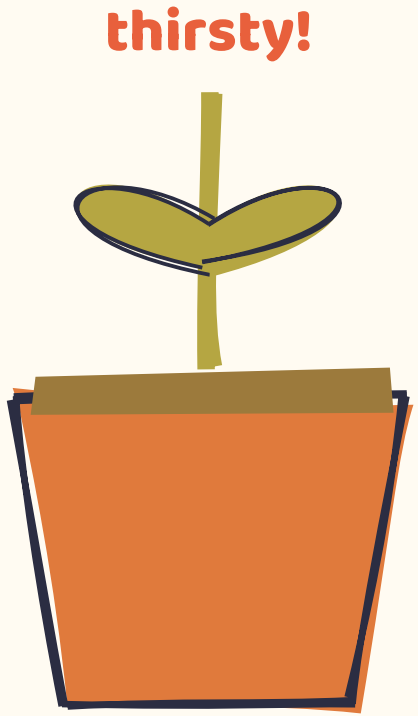
LEARN

Let's wonder...

How can a plant tell us it is thirsty?

Our class plant is sad

Plants cannot talk. They cannot shout **'I need water!'**. By the time the leaves droop, the plant is already very thirsty. How could we know **sooner**?



Ask it?

Touch the soil?

Use a sensor!

Today: we build a **plant alarm** that beeps when the soil is dry!

LEARN

Learning objectives

Today I can...

1	**Plant needs:** name what a plant needs to **live**	**Proof:** tell a friend 3 things plants need
2	**Wet or dry:** use a sensor to tell **wet** soil from **dry** soil	**Proof:** I saw the number change when we added water
3	**Warn us:** make a **beep** and a light when the soil is dry	**Proof:** our plant alarm beeps in dry soil

LEARN

Objective 1 · Plant needs · video

Video time!



The Needs of a Plant for Kids

BuzzWithBee · about 3 minutes · youtu.be/gIRR-VdIP1M

Watch for...

1. What does a plant need to **grow**?
2. Where does a plant drink water from?
3. What happens if a plant gets **no** water?

Scan to watch

After the video: Hold up fingers: how many things does a plant need? Name them with your partner.

Objective 1 · Plant needs

Plants need water, light and air

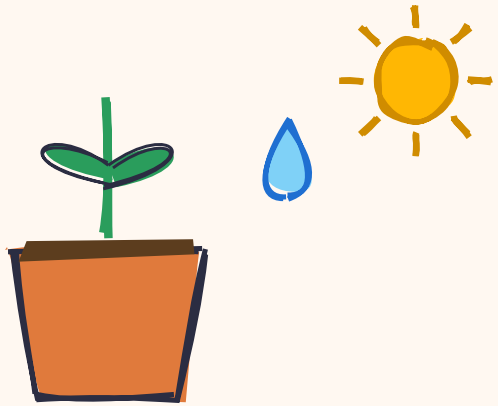
The idea

Plants need **water**, **sunlight**, **air** and good **soil**.

They drink water through their **roots**, under the soil. Too little water and they droop. Too much and the roots get sick!

Plant helpers
Plants also like a little talk and care. Choose a plant helper for each week!

Example: you after playing



After running in the playground on a hot day you feel **thirsty** and tired. A drink of water makes you feel better. Plants feel the same, but they cannot ask!

Objective 2 · Wet or dry

The sensor feels if soil is wet

The idea

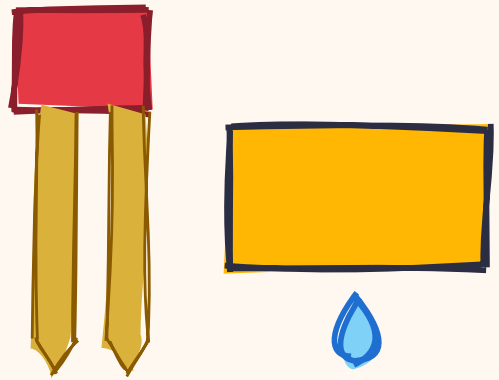
Our sensor has two metal **legs**. We push them into the soil.

Wet soil: electricity passes easily between the legs.
Dry soil: electricity cannot pass well.

The sensor turns this into a **number**. Dry soil and wet soil give **different numbers**. We test both!

Careful
Only the legs go in the soil. Keep the top dry!

Example: a wet sponge



Squeeze a **wet** sponge and water drips out. Squeeze a **dry** sponge and nothing happens. Wet soil lets electricity through, like water through a wet sponge.

Real world

Who helps plants grow?



Farmers
Grow the food we eat: wheat, rice and fruit



Smart sprinklers
Water gardens only when the soil is dry



Greenhouses
Warm glass houses for baby plants



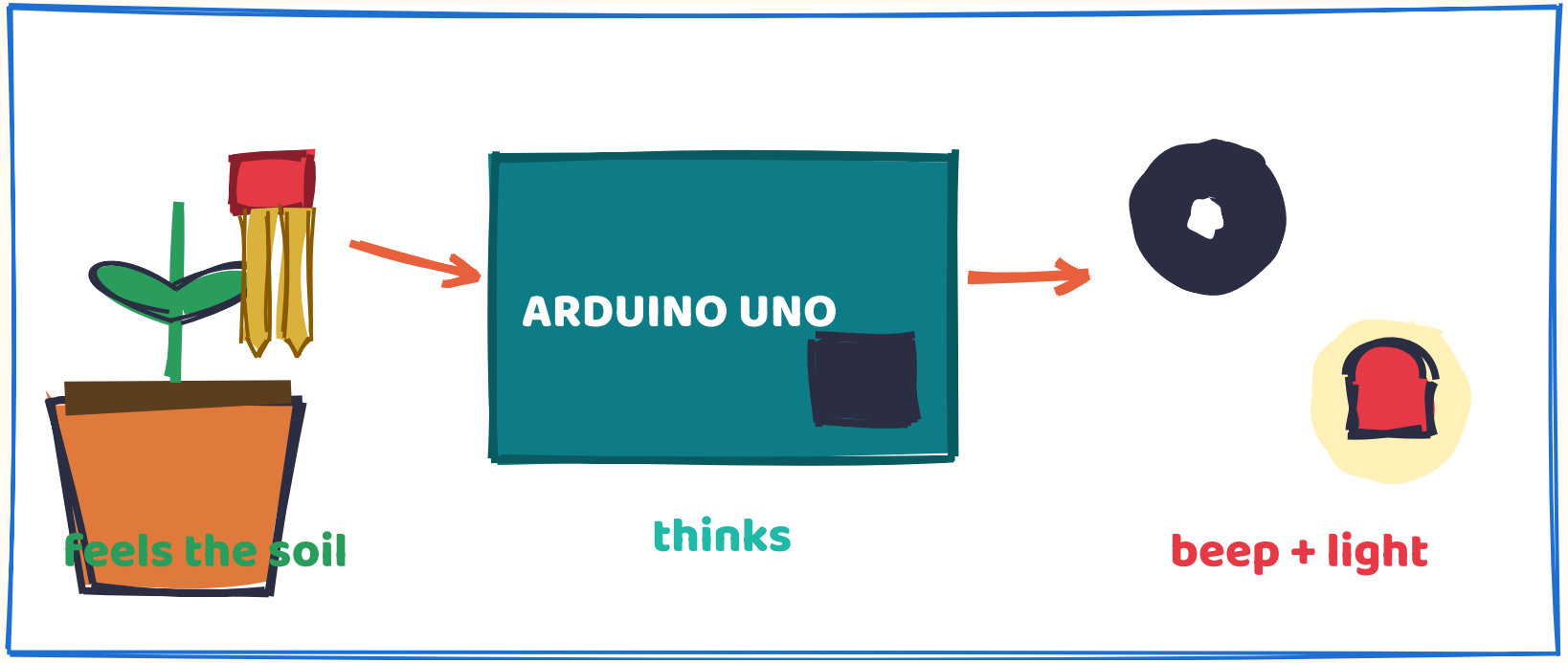
Gardeners
Look after parks and school gardens

? What food did you eat today that came from a plant?

Save water
Watering only when needed saves water for everyone.

Our machine

Feel, think, show



How it works

The **sensor** feels the soil.
The Arduino **thinks**: is the number small (dry)?
Dry? The **buzzer** beeps and the **light** turns on.

Teacher: parts are pre-wired on the breadboard. Children plug in the USB cable, snap the picture blocks in mBlock and test.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Thirsty plant!
2. Snap the picture blocks
3. Dry, damp or wet?
4. Our plant care plan
5. Plant alarm challenges

MAKE

Class game

Thirsty plant!

How to play

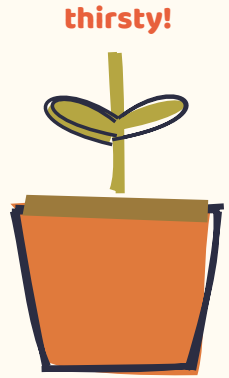
1. Children are plants: crouch down small.
2. The teacher is the sun and the rain.
3. "Sunshine!" Plants stretch up.
4. "Rain!" Plants drink: **slurp!** and grow taller.
5. "No rain for a week!" Plants droop down sadly.

Then discuss

What happens if a plant gets rain **every minute**, all day? Is too much water good?

Big idea

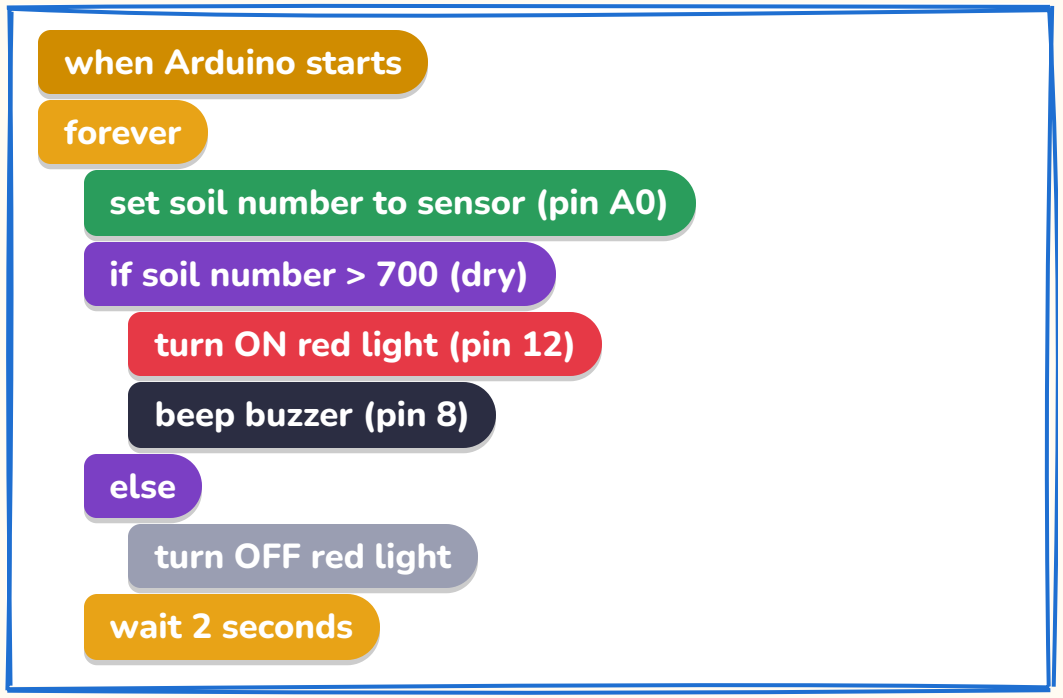
Plants need the RIGHT amount of water: not too little, not too much.



MAKE

Time to build!

Snap the picture blocks



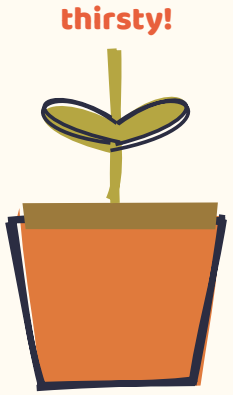
Read it like a story

Read the soil number.

IF it says dry: light and beep!

ELSE: be quiet.

Wait and check again.



Teacher: 700?

Most sensors read about 1000 in dry soil and about 300 in wet soil.
Test yours first and choose a number in between.

MAKE

Experiment!

Dry, damp or wet?

Test	I think it will…	Number we saw
Dry soil	beep / quiet?	_____
Add 1 cup of water	beep / quiet?	_____
Add 2 cups of water	beep / quiet?	_____
Sensor in a glass of water	beep / quiet?	_____

Talk about it

How much water made the alarm **stop**? Was it a little or a lot?

Pour slowly

Pour near the sensor, wait 10 seconds, then read the number.

MAKE

Objective 3 · Warn us · class job

Our plant care plan

Day	Job	Plant helper
Monday	Check the light	_____
Tuesday	Check the light	_____
Wednesday	Check the light	_____
Thursday	Check the light	_____
Friday	Water if it beeps!	_____

Be a scientist

Draw the plant every Friday. Is it **taller**? Does it have **more leaves**?

Do not overwater

Only water when the alarm says dry. Soggy roots get sick!

MAKE

Level up!

Plant alarm challenges

Choose your challenge



Happy light: add a green light when the soil is wet enough.



Thank-you tune: play a happy tune right after we water it.



Too wet!: a different beep when the soil is too wet.



Plant name: give your plant a name and a face card.

Think like an engineer

Which other living things need us to check on them?

Boss level

Make the light blink faster when the soil is VERY dry!

Objective check

Can I really do it? Prove it.

01 Name 3 things a plant needs to live. ○ ○ ○

02 What number did your sensor show in dry soil? And in wet soil? ○ ○ ○

03 Show your alarm beeping in dry soil. ○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Use a dry pot and a wet pot side by side so every child can test.

Show what you know

What did we learn?

- [x] Plants need water, light, air and soil
- [x] The sensor feels if soil is wet or dry
- [x] Dry and wet soil give different numbers
- [x] Our alarm beeps when the plant is thirsty

Exit ticket: what does our alarm do when the soil is dry?

At home

Help water a plant at home. Feel the soil first: dry or wet?



Next class: build a Motion Detector Alarm!